GaitEval: A Multi-Condition Benchmark for Evaluating Whether Large Language Models Use Pose-Derived Gait Evidence

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Abstract

Pose-based gait data are widely used for clinical screening, yet whether Large Language Models (LLMs) can interpret such motion evidence remains underexplored. We introduce GaitEval, a six-dataset benchmark for evidence-grounded label prediction across ASD-Kinect, ATAXIA, CARE-PD, GMFCS, PARKINSONISM, and PATHOGAIT, with heterogeneous populations, pose formats, and label spaces. A baseline-anchored protocol compares controlled pose-derived inputs with label-prior and same-feature linear references, and evaluates zero-shot, few-shot, and low-rank adaptation (LoRA) within their tested coverage. Five-model motion-summary results show limited zero-shot agreement on most tested tasks, while six-dataset Qwen2.5-7B comparisons with source-reported subject-disjoint splits identify important dataset–representation exceptions. Linear references recover task-relevant signal where available, while adaptation recovers some in-distribution mappings: GMFCS summary macro-F1 in the shared-condition evaluation rises from 0.265 to 0.467, but remains below its matched linear reference of 0.579. The ATAXIA adaptation experiment likewise shows an agreement increase from $\kappa = 0.046$ to 0.507; transfer remains unreliable in the tested bidirectional ATAXIA–PATHOGAIT probe. Overall, evidence use is heterogeneous and representation-dependent: adaptation can learn dataset-specific mappings, while reliability beyond the tested settings remains unresolved.

1. Introduction

Human gait provides an accessible signal for assessing balance, coordination, strength, and mobility [1–3]. Advances in video-based pose estimation allow ordinary walking videos to be converted into skeleton trajectories without specialized motion-capture systems [4–6]. Clinical reviews describe how this low-cost route has been used across neurological and rehabilitation settings [7, 8]. Pose trajectories provide a compact, anatomically indexed representation from which interpretable measurements of motion, symmetry, stability, and variability can be derived [9, 10].

Most pose-based clinical gait methods are task-specific supervised systems that map skeleton sequences to gait classes, severity levels, or clinical scores [11, 12]. Clinical studies have related quantitative gait measurements to motor severity and function in cerebral palsy, Parkinsonian impairment, and ataxia [11, 13, 14]. Related visual studies span heterogeneous cohorts, acquisition settings, and label definitions [11, 12, 15], complicating direct cross-resource comparisons. Thus, cross-condition evaluation is not itself the novelty of this work. Instead, we ask whether LLM predictions are grounded in supplied pose-derived evidence. This distinction is important because many gait resources contain numerous clips from relatively few participants, requiring participant-aware interpretation.

Pose evidence can be presented as raw coordinates, deterministic motion summaries, or biomechanical features. Poor performance may therefore reflect limited task-relevant signal, difficulty interpreting the representation, or failure to map informative evidence to dataset-specific labels. Accuracy alone cannot distinguish these cases, particularly for imbalanced ordinal and multi-class tasks [16, 17]. Evaluation should therefore compare LLMs with label-prior baselines and simple classifiers trained on the same evidence.

We introduce **GaitEval**, a multi-condition benchmark in which *gait reasoning* means grounding dataset-specified label predictions in pose-derived evidence, rather than open-ended diagnosis or clinical interpretation. The benchmark comprises six datasets: ASD-Kinect, ATAXIA, CARE-PD, GMFCS, PARKINSONISM, and PATHOGAIT. Each retains its population provenance and native label space. GaitEval introduces no new clinical recordings; it provides controlled representations, evidence-use tests, and same-feature diagnostic references (Fig. 1).

The evaluation follows an evidence ladder: zero-shot prompting, few-shot examples, and LoRA adaptation, with transfer limited to the tested ATAXIA–PATHOGAIT

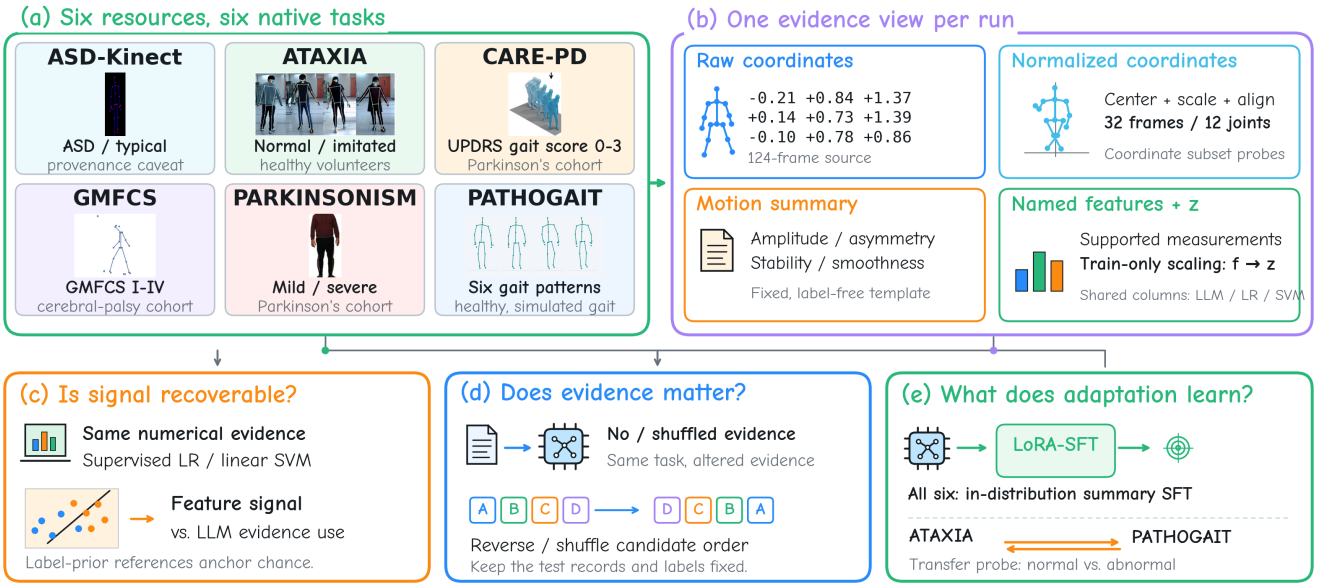


Figure 1. **GaitEval overview.** Six resources retain native tasks. One pose-derived view enters each run; z-scoring is a feature-scaling control. Linear references, evidence/order controls, adaptation, and scoped transfer probe recoverable signal and evidence use. Resource illustrations and coordinate coverage are documented in the supplement.

pair [18, 19]. The common six-dataset analysis uses Qwen2.5-7B-Instruct, source-reported subject-disjoint partitions, and capture-quality-free features. Model, view, and adaptation experiments follow this evaluation protocol, with their actual input settings and eligible test records specified for each comparison.

Our main contributions are:

- We introduce GaitEval, a six-dataset benchmark for testing whether LLM predictions are grounded in pose-derived gait evidence across heterogeneous resources, representations, and label spaces.
- We provide a baseline-anchored protocol that separates limited input signal from failed evidence use by comparing controlled LLM inputs with label-prior and linear feature-based references.
- We evaluate an evidence ladder from zero-shot prompting to few-shot adaptation and LoRA fine-tuning within its measured coverage, showing in-distribution recovery on selected tasks but no reliable transfer on the tested compatible dataset pair.

2. Related Work

2.1. Clinical Gait Analysis and Pose-Based Representations

Clinical gait analysis supports assessment of motor impairment, functional severity, and rehabilitation outcomes [1, 2, 20]. Video-based pose estimation offers an accessible alternative to laboratory motion capture by converting ordinary recordings into joint trajectories [7, 8]. Markerless toolkits and smartphone-based pipelines further broaden access to quantitative movement analysis [21, 22]. Validation studies have compared video-derived temporal and kinematic measures with instrumented walkways or motion-capture systems across different gait patterns and clinical cohorts [23–25]. These studies also show that viewpoint, occlusion, pose-estimation error, and population differences remain important sources of measurement uncertainty.

Gait resources differ in population, acquisition setup, skeleton format, and label definition [11, 13, 15]. Pose data provide a shared semantic interface, but do not make these resources clinically equivalent. GaitEval therefore preserves each resource's native representation and dataset-defined task.

2.2. Visual and Pose-Based Pathological Gait Analysis

Pose-based gait systems use biomechanical features or learned temporal representations for supervised prediction. Graph models such as ST-GCN, 2s-AGCN, and MS-G3D are widely used for skeleton analysis [26–28], but are generally tied to fixed datasets, skeletons, and label spaces.

Early low-cost RGB and RGB-D systems measured or classified pathological gait using semantic gait features, silhouettes, or tracked skeletons [29–31]. Parkinsonian assessment has since ranged from single-camera kinematic measurement and natural-gait monitoring to learned skeleton–silhouette models for clinical severity estimation [32–34]. Related work has also estimated cerebral-palsy gait function and kinematics and quantified gait after stroke from ordinary

videos [13, 35, 36].

For ataxia, Auto-Gait analyzes clinically recorded gait-task videos from multiple sites, while pose-based work has also evaluated performed ataxic gait [11, 37]. A separate clinical study used video keypoints to distinguish Parkinson's disease from spinocerebellar degeneration, showing that cross-condition visual analysis predates GaitEval [38]. Broader abnormal-gait resources include simulated pathological patterns and clinically annotated abnormal-gait videos [15, 39]. Collectively, these studies establish that visual and pose features can support condition-specific prediction; results from performed or simulated gait, however, do not establish patient-level diagnostic validity.

Cross-dataset gait analysis is also established. Transfer-learning and spatiotemporal classifiers have been evaluated across separately collected pathological-gait datasets [40, 41]. Auto-Gait includes leave-one-site-out evaluation, and CARE-PD formalizes within-dataset, cross-dataset, leave-one-dataset-out, and adaptation protocols [37, 42]. GaitEval therefore does not claim domain-shift difficulty as a new finding. Instead, it tests whether general-purpose LLMs use supplied pose evidence under same-input controls. Logistic regression and a linear support vector machine receive the same deterministic features, and the ATAXIA–PATHOGAIT transfer experiment is limited to that compatible dataset pair.

2.3. Language Models for Human Motion and Clinical Gait

Motion-language systems such as MotionGPT, MotionLLM, and MoChat connect language models with motion through learned tokenization or cross-modal alignment [43–45]. Gait-specific models have also been developed for biometric recognition and Parkinsonian scoring [46, 47]. BiomechGPT and BioGait-VLM further connect language-aligned representations with clinically relevant movement analysis [48, 49]. These methods rely on task-specific motion-language training rather than testing off-the-shelf LLM evidence use.

The closest concurrent work evaluates general-purpose LLMs using text-encoded kinematic waveforms [50]. More broadly, prior work has examined LLMs for structured and tabular inputs [51] and multimodal instruction following [52, 53]. GaitEval instead evaluates heterogeneous gait resources through controlled pose-derived views, same-feature linear references, and adaptation experiments. Its purpose is to separate limited motion signal from representation difficulty and failed LLM evidence use, while restricting transfer conclusions to the tested dataset pair.

3. Methods

3.1. Source Sequences and Dataset Tasks

GaitEval evaluates dataset-specified label prediction from pose-derived evidence. The six datasets and their evaluation partitions are listed in Table 1. GMFCS contains children with cerebral palsy [55]; PARKINSONISM and CARE-PD concern Parkinson's disease with different severity or gait-item targets [12, 42]. ATAXIA contains healthy volunteers imitating ataxic gait [56], and PATHOGAIT contains healthy volunteers simulating six gait patterns [15]. ASD-Kinect defines an autism-versus-typical task [67]; its source reports some movements simulated by others without identifying the affected sequences. These resources are therefore not six interchangeable clinical cohorts.

A source sequence is $X_i \in \mathbb{R}^{T_i \times J_d \times C_d}$, with dataset-specific joints and coordinate channels. Native 2D/3D representations follow established pose formats [4, 54]. GaitEval applies one evaluation protocol: select task-eligible participant partitions; derive a label-redacted input view; run the declared model and evidence condition; fit supervised transformations on training data only; and score predictions against the complete task label set. Each experiment records its model, input construction, test IDs, and parser settings. The coordinate-input experiments use a 124-frame source tensor, with padding masked during feature computation; normalized-coordinate prompts sample 32 frames and 12 common joints. Feature and summary runs aggregate the valid trajectories in their recorded input construction.

All six datasets enter the seven-condition Qwen2.5-7B evaluation on the test sets in Table 1. Additional model, coordinate, few-shot, and feature-SFT experiments use the eligible records listed in the supplemental evaluation-record inventory; their test counts are also stated with the results. The GMFCS split-repair check uses 866 test clips from 166 participants; the seven-condition comparison uses 255 clips from 149 participants. Within an experiment, comparisons use matched test records; comparisons across different test populations are descriptive. Participant separation in the 866-clip experiment is independently verified. The seven-condition partitions are reported as subject-disjoint by the source metadata, but complete training manifests remain unavailable for an independent check. Section 4.4 documents the repair.

3.2. Controlled Input Views

Each run exposes one view. Four input-view families are defined: raw coordinates, normalized coordinates, deterministic motion summaries, and named biomechanical features. A train-statistics z-scored encoding is a scaling control within the feature family, not a fifth modality. Coordinate prompting is a subset experiment; it is not claimed to have common six-dataset coverage.

Table 1. **Six benchmark datasets for the seven-condition evaluation (source-reported subject-disjoint partitions).** Counts are eligible train/test clips and test participants, not full-source cohort sizes. All six datasets have summary and z-feature zero-shot, evidence/order controls, summary-SFT, and matched LR/SVM.

Dataset	Population	Task	Train / test	Test subjects
ASD-Kinect	ASD / typical; provenance caveat	Binary	70 / 20	20
ATAXIA	Healthy, imitated gait	Binary	28 / 8	4
CARE-PD	Multi-site Parkinson's cohort	UPDRS gait 0–3	1901 / 665	24
GMFCS	Cerebral-palsy cohort	Four-level ordinal	914 / 255	149
PARKINSONISM	Parkinson's cohort	Binary severity	83 / 29	5
PATHOGAIT	Healthy, simulated gait	Six patterns	5040 / 1440	2

Features describe motion range, bilateral asymmetry, trunk/pelvis variability, trajectory variance, and jerk-based smoothness [9, 10]. Unsupported joints are omitted consistently from the LLM and matched linear inputs. These are pose-coordinate descriptors, not calibrated physical stride lengths or forces. The jerk descriptor uses third finite differences per frame and is not asserted to have SI jerk units. For feature f, scaling is $z = (f - \mu_{\text{tr}})/\sigma_{\text{tr}}$, using training statistics only; constant and missing-column handling is specified in the supplement.

Motion summaries are generated by a deterministic, label-independent template from the numerical measurements, not by annotators, dataset metadata, or another LLM. They preserve numbers and fixed qualitative descriptors. Capture-quality variables, including valid-frame fraction, are excluded from the seven shared feature/summary conditions. Model-visible inputs exclude answer-revealing labels, dataset names, participant IDs, demographics, and clinical metadata; candidate labels remain because they define the task. Figure 2 distinguishes these transformations.

3.3. Models, Adaptation, and Evidence Controls

The five-family motion-summary axis evaluates Llama-3.3-70B, Qwen2.5-72B, and Gemma-3-27B [57–59], together with GLM-4-32B and DeepSeek-R1-Distill-Qwen-32B [60, 61]. The shared six-dataset axis fixes Qwen2.5-7B-Instruct. The test counts for the five-model comparison are 20, 1,647, 665, 866, 120, and 2,181 clips in dataset order; the corresponding Qwen-7B shared-condition counts are in Table 1. Full settings and coverage are in the supplement.

All six datasets use seven common conditions: summary zero-shot; z-feature zero-shot; no evidence; shuffled evidence; reversed and shuffled candidate order; and summary LoRA-SFT. Shuffled evidence deranges summaries across test records while preserving candidate labels. Candidate-order controls preserve evidence and change only label presentation. Decoding is deterministic. The parser chooses the last complete, boundary-anchored candidate mention, resolving same-position matches by length. It does not validate the model's reasoning. Outputs with no matching candidate count as errors.

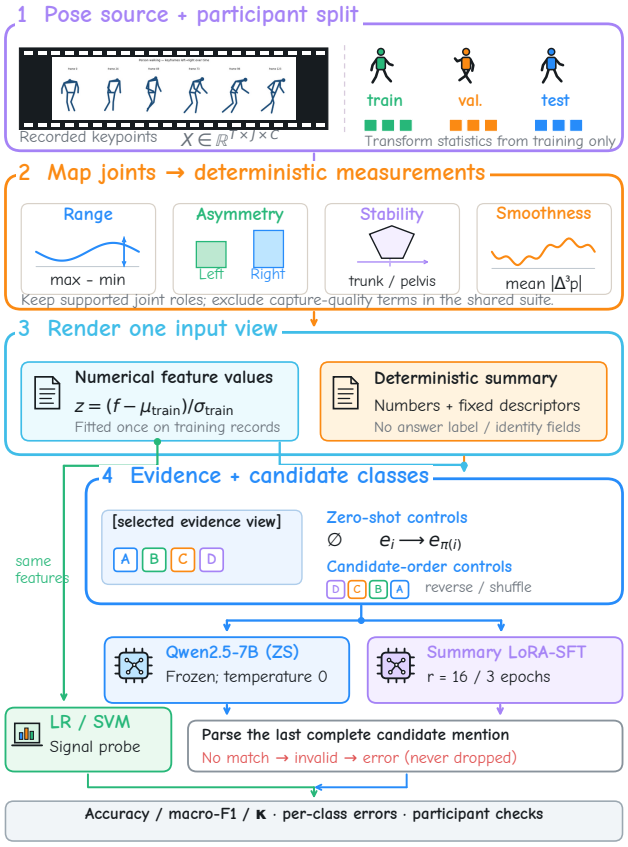


Figure 2. **Pose-to-label evaluation.** The shared six-dataset workflow links training-only transformations to zero-shot controls, summary-SFT, and matched linear references. Recorded GMFCS keyframes illustrate the source; measurement glyphs are schematic. Invalid outputs remain errors. Optional probes follow the supplemental coverage.

Few-shot prompting prepends fixed class-balanced training examples [18]. The tested settings are 8/16/32 examples on ATAXIA, 16 on PARKINSONISM, and 12 on PATHOGAIT; other cells are unexecuted. A reasoning-anchored summary prompt asks for salient motion evidence and candidate comparison [62]; it is also a scoped probe.

LoRA adapts Qwen2.5-7B separately per dataset [19]. The summary-SFT recipe for the seven shared conditions uses rank 16, $\alpha = 32$, dropout 0.05, all linear projections, completion-only loss, three epochs, learning rate 10^{-4}, and effective batch size 16. It saves the final adapter without validation checkpoint selection. No new training is claimed in this revision. Complete prompts, experiment-specific settings, inference budgets, and configuration coverage are given in the supplement.

3.4. References, Metrics, and Scope

Label-prior predictors and class-balanced logistic regression (LR) and linear SVM [63, 64] anchor interpretation. LR/SVM receive the exact deterministic feature columns underlying the corresponding LLM feature view; they are supervised signal probes, not zero-shot competitors or clinical state of the art.

Accuracy, fixed-label-set macro-F1, and Cohen's κ [16, 17, 65] are reported jointly. Here $\kappa = (p_o - p_e)/(1 - p_e)$, with p_e obtained from the empirical target and prediction marginals. Invalid responses remain errors. Ordinal tasks additionally report quadratic-weighted κ and absolute level error. Macro-F1 uses the complete task label set even when a class is unpredicted.

Participant-cluster bootstrap resampling uses 2,000 draws [66]. Historical bootstrap intervals and paired tail areas are identified separately from fixed-label point metrics, since absent classes and an invalid-output sentinel can change the resampled label set. Very small participant counts limit inference regardless of the number of clips. PATHOGAIT's two-participant results are descriptive; no participant-population superiority is claimed. Multiple exploratory contrasts are reported without a family-wide significance claim.

Transfer is measured only on the ATAXIA–PATHOGAIT pair after harmonizing normal versus abnormal gait. A model adapted on one dataset is tested on the other without target labeled examples. This probes combined condition and representation shift; no transfer result is claimed for unmatched severity, gait-score, or ASD targets.

4. Experiments

4.1. Shared Six-Dataset Evaluation

The shared-condition evaluation fixes model, input construction, controls, and eligible test records within each dataset. Table 4 reports all six datasets together. The 42 common cells are backed by individual predictions; two extra raw-feature cells and ten five-family cells provide incomplete additional axes. One archived mixed-task CARE-PD diagnostic is excluded from the common task. No unexecuted cell is interpreted as a failure.

Zero-shot evidence use. Summary zero-shot macro-F1 is near stratified chance or below it on several datasets, but performance alone does not establish whether a model used the input. CARE-PD exceeds shuffled evidence by 0.065 macro-F1 (archived participant-cluster 95% interval $[0.007, 0.118]$, paired bootstrap tail area 0.029). This is evidence of clip-specific responsiveness in that tested setting, while the 0.321 score remains far below its 0.659 LR reference. GMFCS's summary advantage over shuffled evidence is 0.052 (tail area 0.089), and reversing candidate order raises macro-F1 to 0.322. The latter sensitivity cautions against equating plausible accuracy with stable evidence grounding.

Table 2 gives the complete five-family results; Table 3 adds the six-dataset evidence-use contrasts and uncertainty summaries. The five-family axis supports the accepted finding of near-chance agreement in the four larger-test model-comparison experiments ($\kappa = -0.069$ to 0.043 after the GMFCS split correction), but these results concern the stated test populations. The CARE-PD model-comparison cells have $\kappa = 0.099$–0.129; ASD-Kinect ranges from -0.100 to 0.200. Thus, neither the original nor expanded evidence justifies a universal claim that zero-shot LLMs cannot use pose-derived evidence. Original overlapping GMFCS cells are retained only as an explicitly labeled before-correction audit. The 866-clip GMFCS results describe the corrected model-comparison experiment; the shared-condition evaluation separately uses 255 clips.

4.2. Input Representation and Recoverable Signal

Table 5 retains the available input-view measurements and source-reported entries requiring verification, including raw-coordinate subset probes. The shared-condition six-dataset input views and complete linear-reference metrics are reported in Supplementary Tables S29, S30, and S31. They complement the common same-model comparison in Table 4.

Table 5 combines recorded input-view probes rather than a single matched-test ablation. The raw-coordinate row uses Llama-3.3-70B on 150 clips per reported dataset; the other rows use Qwen2.5-7B. GMFCS requires particular care: its summary entry is a log-only 866-clip run, normalized coordinates remain unverified, and both feature entries are historical 899-clip aggregates. These entries are excluded from matched-test comparisons and the following interpretation. The remaining model/view test counts are 20/1,647/665/120/2,181 for ASD-Kinect/ATAXIA/CARE-PD/PARKINSONISM/PATHOGAIT. Input views can change prediction distributions without consistently restoring agreement. For the ATAXIA/PARKINSONISM/PATHOGAIT input-view probes, κ remains between -0.095 and 0.057. In the shared-condition evaluation, PARKINSONISM's named z-feature view reaches 0.757 macro-F1 and $\kappa = 0.520$,

Table 2. **Five-model zero-shot motion-summary benchmark.** Accuracy / macro-F1 / Cohen's κ for ASD-Kinect/ATAXIA/CARE-PD/GMFCS/PARKINSONISM/PATHOGAIT, respectively. These test sets also define the supervision ladder; their sizes differ from the seven-condition comparison where indicated. Bold marks the highest macro-F1 per dataset; invalid outputs count as errors.

Dataset (Maj. Acc.)	**Llama-3.3-70B**	**Qwen2.5-72B**	**Gemma-3-27B**	**GLM-4-32B**	**DeepSeek-R1-32B**
ASD-Kinect (0.500)	0.500 / 0.451 / 0.000	0.450 / 0.373 / −0.100	0.450 / 0.310 / −0.100	0.450 / 0.373 / −0.100	0.600 / **0.583** / 0.200
ATAXIA (0.686)	0.686 / 0.407 / 0.000	0.685 / 0.406 / −0.002	0.686 / 0.407 / 0.000	0.686 / 0.407 / 0.000	0.684 / **0.408** / −0.002
CARE-PD (0.361)	0.441 / 0.262 / 0.119	0.445 / 0.268 / 0.129	0.439 / 0.249 / 0.113	0.441 / **0.350** / 0.124	0.398 / 0.295 / 0.099
GMFCS (0.356)	0.353 / 0.212 / 0.025	0.372 / **0.222** / 0.043	0.313 / 0.163 / 0.030	0.269 / 0.127 / −0.000	0.311 / 0.191 / 0.014
PARKINSONISM (0.583)	0.583 / 0.368 / 0.000	0.575 / **0.472** / 0.035	0.583 / 0.368 / 0.000	0.583 / 0.368 / 0.000	0.583 / 0.368 / 0.000
PATHOGAIT (0.226)	0.159 / 0.062 / −0.010	0.155 / 0.055 / −0.009	0.165 / 0.047 / −0.001	0.141 / 0.069 / −0.069	0.166 / **0.094** / −0.007

Table 3. **Shared six-dataset evidence-use and adaptation evaluation.** Macro-F1 point estimates use the fixed task label set. Bracketed participant-cluster intervals and paired bootstrap tail areas p are archived statistics with the historical resampling convention described in Methods; they are not newly recomputed fixed-label intervals. Δ_{ShufEv} is summary zero-shot minus shuffled evidence; Δ_{SFT} is summary-SFT minus summary zero-shot within Qwen2.5-7B. PATHOGAIT has only two test participants, so its values are descriptive and intervals/p are withheld.

Dataset	Task	Clips (subjects)	Chance	LR [95% CI]	7B ZS [95% CI]	Δ_{ShufEv} (p)	7B SFT (Δ_{SFT}, p)
ASD-Kinect	binary	20 (20)	.490	.697 [.479, .890]	.310 [.200, .394]	−.125 (.25)	.310 (.000, 1.0)
ATAXIA	binary, imitated	8 (4)	.488	.873 [.564, 1.000]	.273 [.111, .333]	.000 (1.0)	.333 (+.061, .60)
CARE-PD	UPDRS gait 0–3	665 (24)	.250	.659 [.407, .742]	.321 [.262, .404]	+.065 (.029)	.445 (+.124, .43)
GMFCS	four-level ordinal	255 (149)	.250	.579 [.474, .662]	.265 [.217, .308]	+.052 (.089)	.467 (+.202, .001)
PARKINSONISM	binary severity	29 (5)	.494	.561 [.416, .654]	.396 [.222, .530]	+.086 (.14)	.542 (+.146, .47)
PATHOGAIT	six-class, simulated	1440 (2)	.166	.881	.069	−.002	.451 (+.382)

versus 0.396 macro-F1 for summaries. Its archived subject-cluster interval is $[0.543, 0.906]$, based on only 29 clips from five participants. The point estimate exceeds the matched linear references, but this small test does not establish population-level superiority. CARE-PD shows the opposite representation preference: summary macro-F1 is 0.321 versus 0.133 for z-features.

On ASD-Kinect, summary and z-feature scores are 0.310 and 0.436, both below the matched linear point estimates. ATAXIA's eight clips cannot support a powered model ranking. PATHOGAIT is low under both zero-shot views; GMFCS summaries outperform its z-feature encoding. These six rows jointly show representation dependence rather than a universally best input view.

The same-feature references establish recoverable supervised signal, with LR macro-F1 from 0.561 to 0.881 in the shared-condition evaluation. Reference strength and uncertainty remain dataset-specific: ASD-Kinect LR has a wide interval and an archived permutation value of 0.09. They do not show that supervised and zero-shot learning have equivalent information access. Per-class recall and confusion matrices in the supplement expose concentrated outputs that aggregate accuracy can conceal.

4.3. Adaptation Recovers Dataset-Specific Mappings

Summary-SFT changes macro-F1 by 0.000, 0.061, 0.124, 0.202, 0.146, and 0.382 for ASD-Kinect, ATAXIA, CARE-PD, GMFCS, PARKINSONISM, and PATHOGAIT, respectively. GMFCS improves to 0.467; its archived paired participant-bootstrap tail area is 0.001. CARE-PD improves to 0.445, but its corresponding tail area is 0.43. Both remain below matched LR. PARKINSONISM SFT remains below its zero-shot z-feature result. PATHOGAIT improves substantially but remains below its capture-quality-free linear references; two participants prevent a population-level gain claim. ASD-Kinect shows no summary-SFT macro-F1 gain, despite a change in the class receiving most predictions.

The adaptation ladder reports all available supervision settings on their stated test records (Supplementary Table S32). Few-shot ATAXIA improves from 0.406 to 0.697 macro-F1. Within the same 7B model, ATAXIA summary agreement rises from $\kappa = 0.046$ to 0.507, and feature-SFT reaches 0.762 macro-F1 versus LR 0.763. Its participant comparison does not detect a difference ($p = 0.97$); this is not an equivalence test. PATHOGAIT feature-SFT reaches 0.687 versus LR 0.662, but the direction reverses for one of its two participants. These results support in-distribution recovery on the tested populations; they do not establish general diagnostic knowledge.

One PATHOGAIT feature in the 2,181-clip adaptation experiment, valid-frame fraction, carries capture-related information. Removing it reduces LR/SVM macro-F1 from 0.662/0.644 to 0.564/0.560. A recovered feature-SFT rerun also removes that field and reaches 0.632 macro-F1, compared with 0.687 when it is retained. The seven shared con-

Table 4. **Same-model evidence use and adaptation across all six datasets.** Values are macro-F1, recomputed from released predictions with a fixed task label set. LR/SVM values are the committed task-matched supervised references. N is clips/participants. Best LLM view is bold; bold does not imply superiority over references.

Dataset	N	Chance	ZS summary	ZS z-features	Summary SFT	LR	SVM
ASD-Kinect	20/20	.490	.310	**.436**	.310	.697	.744
ATAXIA	8/4	.488	.273	**.333**	**.333**	.873	.873
CARE-PD	665/24	.250	.321	.133	**.445**	.659	.650
GMFCS	255/149	.250	.265	.150	**.467**	.579	.571
PARKINSONISM	29/5	.494	.396	**.757**	.542	.561	.574
PATHOGAIT	1440/2	.166	.069	.139	**.451**	.881	.897

Table 5. **Four input-view families and a feature-scaling control.** Accuracy / macro-F1 / κ. Z-scoring is an encoding of the named features. "–" means no recoverable numeric result.

Input view	ASD-Kinect	ATAXIA	CARE-PD	GMFCS	PARKINSONISM	PATHOGAIT
Raw coordinates[†]	–	0.573 / 0.323 / 0.009	–	–	–	0.093 / 0.083 / 0.038
Normalized coordinates	–	0.316 / 0.244 / 0.002	–	0.282 / 0.110 / 0.000	0.442 / 0.441 / −0.095	0.110 / 0.034 / 0.000
Motion summary	0.450 / 0.310 / −0.100	0.692 / 0.452 / 0.046	0.433 / 0.321 / 0.152	0.259 / 0.113 / −0.009	0.550 / 0.400 / −0.045	0.142 / 0.087 / −0.024
Named feature vector	0.500 / 0.333 / 0.000	0.382 / 0.347 / 0.057	0.332 / 0.221 / 0.039	0.343 / 0.245 / 0.007	0.417 / 0.294 / 0.000	0.110 / 0.033 / 0.000
Z-scored named feature vector	0.550 / 0.436 / 0.100	0.583 / **0.495** / −0.008	0.233 / 0.133 / 0.020	0.185 / 0.100 / −0.049	0.508 / **0.493** / −0.014	0.150 / **0.130** / −0.011

ditions exclude capture-quality variables for all six datasets. Their PATHOGAIT test has 1,440 clips, so the 2,181-clip feature-removal check is interpreted on its own matched test IDs. We therefore interpret adaptation as learning from the supplied measurements, without a kinematics-only or population-level superiority claim.

4.4. Corrected Splits and Feature-Quality Checks

GMFCS participant separation. The original partition split visits rather than participants: three patients occurred in both training and test, affecting 14 of 899 test clips. Checking validation as well identifies additional overlap. We recover the released deterministic repair, which assigns each participant to the first partition in which they occur, in training–validation–test order, and moves all their clips together. This produces 1,602 training, 478 validation, and 866 test clips from 94, 84, and 166 participants, respectively. Direct comparison of the released participant maps gives empty intersections for all three partition pairs. Each new index retains its original split and sample index, making the 33 removed test clips traceable rather than silently discarded.

This repair is followed by training-dependent reruns, not solely by deleting overlapping test clips from an already trained SFT evaluation. The archived linear references and 7B summary-SFT are retrained on the repaired training partition. Their released SFT predictions contain exactly the 866 corrected test IDs, with labels agreeing with the original records. Re-parsing the raw generations gives 0.3730 accuracy, 0.2585 macro-F1, and $\kappa = 0.0431$, with no invalid responses. The archived balanced LR/SVM macro-F1 values are 0.3952/0.4121; the SFT score remains lower. Supplementary Table S35 distinguishes the original and corrected experiments. The original and corrected SFT columns use different training and test sets, so their small aggregate difference is not a paired estimate of the effect of leakage.

The five zero-shot motion-summary models have no fitted training stage on this task. Their cached raw outputs can therefore be restricted to the corrected test IDs and re-parsed without retraining a model. The archived corrected macro-F1 values are 0.2122, 0.2216, 0.1634, 0.1266, and 0.1905 for Llama, Qwen, Gemma, GLM, and DeepSeek, respectively. These remain below the corrected stratified-random reference of 0.2498; the largest corrected κ is 0.0426. Thus, the repair changes the recorded cells and uncertainty analysis while retaining the finding of limited zero-shot agreement in this protocol. It does not support the earlier, stronger interpretation that every model produces exactly the same severity level.

Evaluation records. The split-repair experiment has 866 test clips from 166 participants; the seven-condition GMFCS evaluation has 255 clips from 149 participants and 16 supported feature columns. LR macro-F1 is 0.3952 on the former and 0.5788 on the latter. These values refer to different test populations, not repeated estimates on identical records. Within-experiment comparisons retain their own IDs and feature columns. Verification of the repaired participant maps does not certify the unavailable complete training manifest for the 255-clip evaluation.

PATHOGAIT without valid-frame fraction. The released ablation deletes only the named valid-frame-fraction line from the 2,181-clip experiment's feature representation before constructing SFT examples. Its generator asserts that

Table 6. **Bidirectional ATAXIA–PATHOGAIT transfer.** Models use z-scored feature descriptions; target-majority values are accuracy / macro-F1. Target tests have 2,181 PATHOGAIT clips or 1,647 ATAXIA clips. PATHOGAIT is binarized, so its in-distribution κ is not comparable with the six-class adaptation result. OOD denotes target-dataset evaluation. Only this task-compatible pair was tested.

Train → Test	In-dist. κ	OOD Acc. / mF1 / κ	Target majority
ATAXIA → PATHOGAIT	0.549	0.656 / 0.496 / 0.057	0.890 / 0.471
PATHOGAIT → ATAXIA	0.917	0.686 / 0.407 / 0.000	0.686 / 0.407

the field was removed from every record and leaves IDs, labels, record order, and the remaining feature text unchanged. We compare the archived original and ablated predictions on identical sets of 2,181 test IDs and true labels, using the same six-label parser and scoring rule. Supplementary Table S37 reports the recomputed SFT results; the corresponding archived linear-reference comparisons are documented in the supplementary feature audit. The ablated SFT model reaches 0.6410 accuracy and $\kappa = 0.5649$, with no invalid predictions.

Removing the field reduces SFT macro-F1 by 0.0554, while the resulting point estimate remains above the quality-free linear references. This establishes that a dataset-specific mapping can still be learned without that field in the recorded rerun. It does not establish a general advantage: PATHOGAIT has two test participants, the rerun is a single training realization, and the source log records a change of serving/training environment. We do not attribute the entire difference to the removed field. The seven shared conditions independently exclude capture-quality variables; feature-SFT on their 1,440-clip PATHOGAIT test set has not been run.

4.5. Transfer Across the Tested Dataset Pair

Table 6 retains the complete transfer and target-majority results. In the tested ATAXIA–PATHOGAIT transfer probe, in-domain κ is 0.549 and 0.917 for the two directions, while target-domain κ is 0.057 and 0.000. PATHOGAIT is binarized to an approximately 11%/89% target distribution; the resulting binary accuracy cannot be compared with six-class scores. This supports only the accepted conclusion that reliable transfer was not observed on this harmonized pair. Six-dataset membership does not imply task-compatible transfer pairs.

4.6. Release and Limitations

Supplementary Table S23 records completed and missing configurations across the same six benchmark members. The same dataset membership applies to every experiment; model and input coverage are stated explicitly. Missing configurations are not interpreted as zero performance.

The independent project repository provides six-dataset predictions, scoring/plotting code, runner and SFT configurations, table-to-file mappings, checksums, and source-access notes. The correction bundle adds the GMFCS participant maps for the corrected 866-clip experiment and retrained-model predictions, PATHOGAIT feature-removal predictions, and scoring evidence. These assets originate from the authors' released GaitEval repository and the source camera-ready archive, with immutable commits recorded in the bundle. They support score and split checks without a new model execution. Source poses, complete shared-condition input layers, adapters, and full training logs are not bundled, so end-to-end retraining still requires authorized external inputs. Dataset and image rights remain separate from the code license. Original RGB recordings are not redistributed; GMFCS source access concerns de-identified trajectories rather than clinical video. The permission ledger distinguishes source datasets, publication illustrations, and newly rendered skeletons. A dataset access link or code license alone is not treated as permission to redistribute recordings or published illustrations.

Small test populations, performed-gait provenance, source simulation uncertainty, unequal class support, and missing training artifacts limit interpretation even when the number of clips is large. The verified 866-clip split repair answers the original GMFCS overlap concern for that experiment; it does not independently certify the 255-clip evaluation's training partition. The two-participant PATHOGAIT ablation is a fixed-split observation, not evidence of population-level superiority. This revision incorporates existing reruns and checks their released outputs; no new model training or inference is claimed. Complete shared-condition inputs and unresolved source-image reuse permissions remain listed in the release checklist.

5. Conclusion

We introduced GaitEval, a six-dataset benchmark for grounding dataset-specified gait-label predictions in pose-derived motion evidence. Controlled inputs and same-feature linear references distinguish limited signal from failed evidence use. The five-family and same-model results retain the accepted finding of limited zero-shot agreement in many tested settings, with dataset–representation exceptions under the shared six-dataset protocol. Task-specific adaptation recovers selected in-distribution mappings; the retained ATAXIA/PATHOGAIT adaptation results do not establish clinical or population-level superiority. Reliable transfer is not observed in the tested bidirectional ATAXIA–PATHOGAIT probe. Overall, GaitEval shows that LLM grounding in pose-derived gait evidence is heterogeneous and representation-dependent: adaptation can recover dataset-specific mappings in distribution, but reliability beyond the tested settings remains unresolved.

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